Discussion Point:

Which network topologies have you used? You might wish to defer this discussion until you have read about network devices later in this chapter.

In a situation where several LANs are connected, they can have different topologies or supporting technologies. This collection of LANs then becomes a **hybrid network**. A special connecting device is needed to ensure that the hybrid network is fully functional. It is often an advantage to be able to connect a new topology LAN to existing LANs where it is not sensible or not possible to use the existing topology for the new LAN. An example is when a wired LAN is already installed but a new wireless LAN is to be connected to it.

KEY TERMS

Hybrid network: a collection of connected LANs where some of them have different topologies or supporting technologies

Cable: a transmission using copper wire or fibre-optic

Bandwidth: a measure of the amount of data that can be transmitted per second

2.03 Transmission media

Cable

A network **cable** can be twisted pair, coaxial or fibre-optic. The twisted pair and coaxial cables both use copper for the transmission medium. In discussing suitability for a given application there are a number of factors to consider. One of these is the cost of the cable and connecting devices. Another is the best **bandwidth** that can be achieved. The bandwidth governs the possible data transmission rate. There are then two factors that can cause poor performance: the likelihood of interference affecting transmitted signals and the extent of attenuation (deterioration of the signal) when high frequencies are transmitted. These factors will dictate whether repeaters or amplifiers are needed in transmission lines and how many will be needed. Table 2.01 shows some comparisons of the different cable types.

	Twisted pair	Coaxial	Fibre-optic
Cost	Lowest	Higher	Highest
Bandwidth or data rate	Lowest	Higher	Much higher
Attenuation at high frequency	Affected	Most affected	Least affected
Interference	Worst affected	Less affected	Least affected
Need for repeaters	More often	More often	Less often

Table 2.01 Comparisons between cable types

You need to understand that for each of the three types of cabling there are defined standards for different grades of cable which must be considered when you decide which type of cable to use. Fibre-optic cable performs best but costs more than the other kinds. For a new installation the improved performance of fibre-optic cable is likely to be the factor that governs your choice. However, where copper cable is already installed the cost of replacement by fibre-optic cable may not be justified.

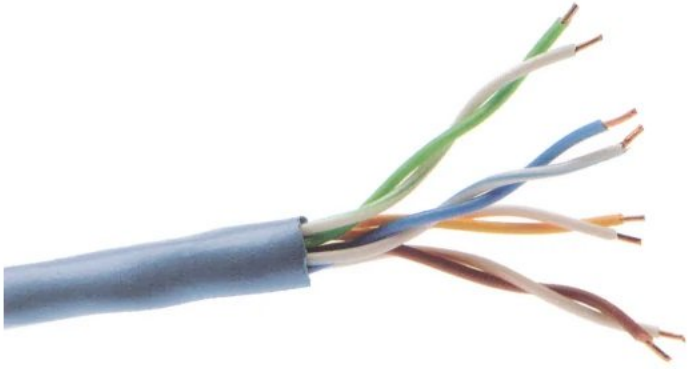


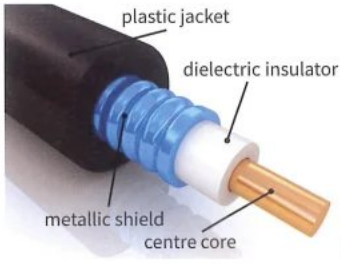
Figure 2.05 One cable with four twisted pairs with differing twist rates to reduce interference

Currently, twisted pair cable is normally used to connect telephone handsets to telephone lines. This type of cable is illustrated in Figure 2.05. It is also the technology of choice for high-speed local area networks.

Question 2.01

Twisted pair cable can be shielded or unshielded. What are the options for this? How does shielding affect the use of the cable?

Coaxial cable is used extensively by cable television companies and in metropolitan area networks. It is not usually used for long-distance telephone cabling. Fibre-optic cable is the technology of choice for long-distance cabling. As shown in Figure 2.06, coaxial cable is not bundled but a fibre-optic cable contains many individual fibres.



a



b

Figure 2.06 (a) Coaxial cable and (b) a bundled fibre-optic cable

Wireless

The alternative to cable is **wireless** transmission. The three options here are radio, microwave or infrared. These are all examples of electromagnetic radiation; the only intrinsic difference between the three types is the frequency of the waves.

KEY TERM

Wireless: a transmission using radio, microwave or infrared

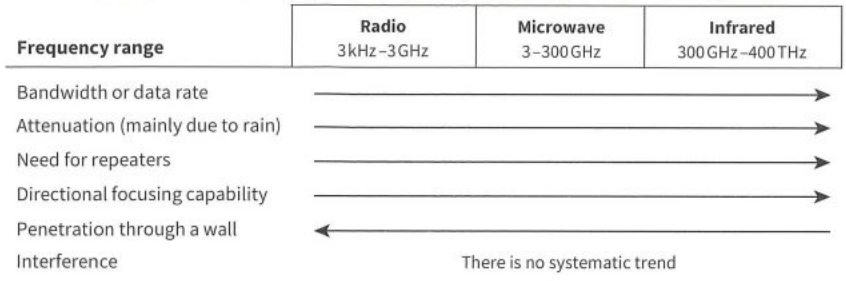


Figure 2.07 Frequency ranges and frequency dependency of factors affecting wireless transmission

When making a choice of which wireless option to use you need to consider all of the same factors that were discussed when comparing different kinds of cable. In addition, the ability of the radiation to transmit through a solid barrier is an important factor. Also, the extent to which the transmission can be focused in a specific direction needs to be considered. Figure 2.07 shows the approximate frequency ranges for the three types of radiation. The factors listed on the left increase in the direction of the arrows. The bandwidth increases through radio and microwave to infrared but the ability of the waves to penetrate solid objects is greatest for radio waves. Interference is not consistently affected by the frequency.

The increased attenuation for infrared transmission, which has the highest frequency, means that it is only suitable for indoor applications. The fact that it will not penetrate through a wall is then of benefit because the transmission cannot escape and cause unwanted interference elsewhere. For most applications, microwave transmission is the best option because it has a better bandwidth compared to that available using radio waves.

Comparing cable and wireless transmission

It is worth noting that cables are often referred to as 'guided media' and wireless as 'unguided media'. This is slightly misleading because only radio wave transmission fits the description of unguided. It is possible with microwaves or infrared to direct a transmission towards a particular receiver (as suggested in Figure 2.07).

There are other points to consider when we compare the relative advantages of transmission through a cable or wireless transmission.

- The use of certain wireless transmission frequencies is regulated by government agencies and so permission has to be obtained before wireless transmission is used.
- Outside these frequencies, no permission is needed to use the air for transmission but cables can only be laid in the ground with the permission of landowners.
- For global communications, the two competing technologies are: transmission through fibre-optic cables laid underground (or on the sea bed) and satellite transmission (discussed later in this section).
- Interference is much more significant for wireless transmission and its extent is dependent on which frequencies are being used for different applications.
- Repeaters are needed less often for wireless transmission.
- Mobile (cell) phones now dominate Internet use and for these, only wireless transmission is possible.
- For home or small office use, wired or wireless transmission is equally efficient; often, not having to install cables favours wireless connections for a small network.

Satellites are components of modern communication systems. Figure 2.08 shows the altitudes (distances above Earth) of three different types of satellite. The Van Allen belts are areas containing high levels of electrically charged particles, which interfere with satellites.

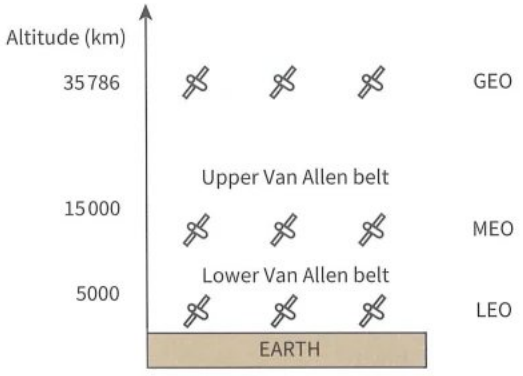


Figure 2.08 Satellite altitudes

The highest altitude satellites are in geostationary Earth orbit (GEO) over the equator and these are used to provide long-distance telephone and computer network communication. 'Geostationary' means that the satellite orbits at the same speed as the Earth spins, so from a point on the Earth the satellite always appears to be at the same point in the sky. Only three GEO satellites are needed for full global coverage. Closer to Earth are a group of medium-Earth-orbit (MEO) satellites some of which provide the global positioning system (GPS). Ten MEO satellites are needed for global coverage. Finally, low-Earth-orbit (LEO) satellites work in 'constellations' to supplement the mobile phone networks. Fifty LEO satellites are needed for full global coverage but currently there are several hundred LEO satellites in orbit.

A satellite can act as a component in a network and can directly connect with ground-based components. These ground-based components can be much further apart than in a network with no satellites. The disadvantage of satellites is that the greater transmission distance causes transmission delays, which can cause technical problems for the network.

TASK 2.01

Calculate the approximate time taken for a transmission from the surface of the Earth to a medium-Earth-orbit satellite. (Take the speed of light to be 300 000 km per second.)

The use of satellites in networks tends to be for specialised applications such as the Global Positioning System (GPS) or for Internet use in remote locations. At one stage, a lot of Internet communication was expected to make use of satellites, but the development of high-speed fibre-optic cabling at relatively low cost has reduced the need for satellites.

2.04 LAN hardware

Wired LANs

In the early years, coaxial cable was used for LANs. Nowadays, twisted pair cables are probably the most widely used networking connections, and fibre-optic cables are becoming more common. In a bus configuration the bus will consist of a series of sockets linked by cables. The ends of the bus have terminators attached that prevent signals from reflecting

back down the bus. Each end-system (which is either a user workstation or a **server**), has a short length of cable with an RJ-45 connector at each end. One end is plugged into a bus socket and the other end is plugged into the LAN port of the end-system.

In a star configuration each end-system has the same type of cable with the same connectors but the cable tends to be much longer because it has to plug into a socket on the central device.

A bus can be extended by linking two bus cables using a **repeater**. A repeater is needed because over long distances, signals become attenuated (reduced in strength), making communication unreliable. A repeater receives an input signal and generates a new full-strength signal. Sometimes a bus network is constructed in what are called segments. Two segments are connected using a **bridge**. The bridge stores the network addresses for the end-systems in the two segments it connects.

The LAN port on an end-system is connected to a **Network Interface Card (NIC)**. The NIC is manufactured with a unique network address that is used to identify the end-system in which it has been installed. The addressing system is discussed in Chapter 17 (Section 17.05). For a star network, the central device might be a hub, a **switch** or a router. The switch is by far the most likely. A switch is a connecting device that can direct a communication to a specific end-system. There is discussion of how it functions in Section 2.05. The router is discussed later in this chapter and also in Chapter 17.

Wireless LANs

WiFi (WLAN in some countries) is a term used to describe wireless Ethernet. Its formal description is IEEE 802.11. This is a wireless LAN standard that uses radio frequency transmission. The central device in a WiFi LAN is a **Wireless Access Point (WAP)**. This can be an end-system in a wired network. The WAP can communicate with an end-system in the WiFi LAN provided that the end-system has a **Wireless Network Interface Card (WNIC)** installed.



KEY TERMS

Server: a system providing a service to end-systems
Repeater: a device that connects two cables and provides a full-strength signal to the second cable
Bridge: a device that connects two segments of a LAN
Network Interface Card (NIC): a component used to identify the end-system
Switch: a connecting device that can send a unicast message
Wireless Access Point (WAP): the connecting device in a WiFi LAN
Wireless Network Interface Card (WNIC): provides the NIC function in a WiFi LAN

2.05 Ethernet

Ethernet is one of the two dominant technologies in the modern networked world. It is primarily focused on LANs. Although Ethernet was first devised in the 1970s independently of any organisation, it was later adopted for standardisation by the Institute of Electrical and Electronics Engineers (IEEE). In particular it was their 802 committee that took responsibility for the development of the protocol. The standard for a wired network is denoted as IEEE 802.3 which is sometimes used as an alternative name for Ethernet. The standard has so far evolved through five generations: standard or traditional, fast, gigabit, 10 gigabit and 100 gigabit. The gigabit part of the name indicates its data transfer speed capability.

Original (or 'legacy') Ethernet was implemented on a LAN configured either as a bus or as a star with a hub as the central device. In either topology, a transmission was broadcast type. Any message would be made available to all of the end-systems without any controlled

communication exchange between any pair of end-systems. For each message received an end-system had to check the destination address defined in the message to see if it was the intended recipient.

The use of a shared medium for message transmission has the potential for messages to be corrupted during transmission. If two end-systems were to transmit messages at the same time there would be what is described as a 'collision'. This is when the voltages associated with the transmission interfere with each other causing corruption of the individual messages. The method adopted for dealing with this was CSMA/CD (carrier sense multiple access with collision detection). This relied on the fact that if a message was being transmitted there was a voltage level on the Ethernet cable which could be detected by an end-system.

The transmitter uses the following procedure.

1. Check the voltage on the transmission medium.
2. If this indicates activity, wait a random time before checking again.
3. If no activity is detected, start transmission.
4. Continuously check for a collision.
5. If no collision is detected, continue transmission.
6. If a collision is detected, stop transmission of the message and transmit a jamming signal to warn all end-stations; after a random time, try again.

Although there might be some legacy Ethernet LANs still operating, modern Ethernet is switched. The star configuration has a switch as the central device. The switch controls transmission to specific end-systems. Each end-system is connected to the switch by a full-duplex link, so no collision is possible along that link. Because there might be high levels of activity the switch needs to be able to store an incoming message in a buffer until the cable is free for the transmission to take place. Since collisions are now impossible, CSMA/CD is no longer needed. Some further details concerning Ethernet are provided in Chapter 17 (Section 17.04).

Discussion Point:

Carry out some research about the different versions of Ethernet. Which version is implemented for the systems you use? For how long will it give sufficient performance?

2.06 The Internet infrastructure

To describe the Internet as a WAN pays little attention to its size and complexity. The Internet is the biggest internetwork in existence. Furthermore, it has never been designed as a single 'whole'; it has just evolved to reach its current form and is still evolving towards whatever future form it will take.

Internet Service Provider (ISP)

One of the consequences of the Internet not having been designed is that there is no agreed definition of its structure. However, there is a hierarchical aspect to the structure (meaning that there are several distinct 'levels' within the structure). For example, the initial function of an Internet Service Provider (ISP) was to give Internet access to an individual or company. This function is now performed by what we can call an 'access ISP'. These access ISPs then connect to what we can call 'middle tier' or regional ISPs, which in turn are connected to tier 1 (or 'backbone') ISPs. An ISP is a network and connections between ISPs are handled by Internet Exchange Points (IXPs). The tier 1 ISPs are at the top of the hierarchy, along with major Internet content providers.

Discussion Point:

How many ISPs or major Internet content providers are you familiar with?

Router

We can also think of the Internet in terms of the connections that carry the most traffic, which consist of a set of fibre-optic cables laid under the sea and across land, which can be described as a 'mesh' structure. This mesh of cables contains many points where the cables connect together, which we call nodes. At every node is a device called the **router**. Routers are found not only in the general 'mesh' of the Internet but also within the ISP networks. Each router is connected to several other routers and its function is to choose the best route for a transmission. The details of how a router works are discussed in Chapter 17 (Section 17.05).

KEY TERM

Router: a device that acts as a node on the Internet

Question 2.02

How near are you to an under-the-sea Internet fibre-optic cable?

Public switched telephone network (PSTN)

Communication systems that were not originally designed for computer networking provide significant infrastructure support for the Internet. The longest standing example is what is often referred to as POTS (plain old telephone service) but is more formally described as a PSTN (public switched telephone network). There is some discussion about how PSTNs provide that support in Chapter 17. During the early years of networking the telephone network carried analogue voice data. However, digital data could be transmitted provided that a modem was used to convert the digital data to analogue signals. Another modem was used to reverse the process at the receiving end. Such so-called 'dial-up' connections provided modest-speed, shared access when required. However, an organisation could instead pay for a leased line service that provided a dedicated, permanently connected link with guaranteed transmission speed. Typically, organisations made use of leased lines to establish WANs (or possibly MANs (metropolitan area networks)).

More recently, the PSTNs have upgraded their main communication lines to fibre-optic cable employing digital technology. This has allowed them to offer improved leased line services to ISPs but has also given them the opportunity to provide their own ISP services. In this role they provide two types of service. The first is a broadband network connection for traditional network access. The second is WiFi hotspot technology, where an access point as described in Section 2.04 has a connection to a wired network providing Internet access.

Cell phone network

For users of devices with mobile (cell) phone capability there is an alternative method for gaining Internet access. This is provided by mobile phone companies acting as ISPs. The mobile phone, equipped with the appropriate software, communicates with a standard cell tower to access the wireless telephone network, which in turn provides a connection to the Internet.

2.07 Applications that make use of the Internet

The World Wide Web (WWW)

It is common practice to talk about 'using the web' or 'using the Internet' as though these were just two different ways of saying the same thing. This is not true. The Internet is, as has been described above, an Internetwork. By contrast, the World Wide Web (WWW) is a distributed application which is available on the Internet.

Specifically, the web consists of an enormous collection of websites each having one or more web pages. The special feature of a web page is that it can contain hyperlinks which, when clicked, give direct and essentially immediate access to other web pages.

Cloud computing

Cloud computing is the provision of computing services usually via the Internet. An organisation may choose to establish its own **private cloud**. In this case there are three possible approaches:

- The organisation takes full responsibility for creating and managing the cloud installed on-site and connected to a private network
- The organisation outsources to a third-party the creation and management of an on-site installation connected to a private network
- The organisation outsources the creation and management of an Internet accessible system by a third-party.

The alternative is a **public cloud**. This is created, managed and owned by a third-party cloud service provider.

The services provided by a cloud are familiar ones provided by file servers and application servers. They are accessible via a browser and therefore accessible from any suitable device in any location. A public cloud can be accessed by an individual user or by an organisation. One major difference is the scale of the systems. The provision is established using large mainframe computers or server farms. The services provided can be characterised as being one of:

- infrastructure provision
- platform provision
- software provision

Many of the advantages to a cloud user arise from the fact that the cloud does not have the limitations that the systems already available have. For the infrastructure provision, the advantages include the better performance when running software and the increased storage capacity. For the platform provision, the cloud can offer facilities for software development and testing. For the software provision, the cloud will be able to run applications that require high performance systems. Alternatively, it could be that the costs to a company of buying and installing a software package themselves would be far too high. The other advantage is the familiar one with regard to outsourcing. The cloud user no longer needs technical expertise.

The disadvantages to a cloud user relate to the use of a public cloud. The cloud service provider has complete access to all of the data stored on the cloud. The cloud user cannot be sure that their data is not being shared with third-parties. This is a concern with regard to data privacy. The security of the data stored is an issue; the cloud service provider is being relied on to ensure data cannot be lost.



KEY TERMS

Private cloud: owned by and only accessed by an organisation

Public cloud: owned by a cloud service provider for general access

Bit streaming

Streaming media make use of the Internet for leisure activities like listening to music or watching a video. But what is a 'bit stream'? In general, before data is transmitted it is stored in bytes which can be transmitted one after the other as a 'byte stream'. Because of the file sizes involved, streamed media is always compressed to a sequence of bits - a 'bit stream'. Generic compression techniques mentioned in Chapter 1 (Section 1.07) can convert the byte stream to a bit stream with fewer bits overall. For the decoding process at the receiver end to work properly, the data must be transferred as a bit stream.

For one category of streaming media, the source is a website that has the media already stored. One option in this case is for the user to download a file then listen to it or watch it at some future convenient time. However, when the user does not wish to wait that long there is the streaming option. This option is described as viewing or listening **on demand**. In this case the delivery of the media and the playing of the media are two separate processes. The incoming media data are received into a buffer created on the user's computer. The user's machine has media player software that takes the media data from the buffer and plays it.

The other category of streaming media is **real-time** or live transmission. In this case the content is being generated as it is being delivered such as when viewing a sporting event. At the receiver end the technology is the same as before. The major problem is at the delivery end because a very large number of users may be watching simultaneously. The way this is managed now is to transmit the media initially to a large number of content provider servers which then transmit onwards to individual users.

A crucial point with media streaming is whether the technology has sufficient power to provide a satisfactory user experience. When the media is created it is the intention that the media is to be delivered to the user at precisely the same speed as used for its creation; a song that lasted four minutes when sung for the recording would sound very peculiar if, when it is received by a user, it lasts six minutes. The process of delivering the content is determined by the **bit rate**. For example, a relatively poor-quality video can be delivered at a bit rate of 300 kbps but a reasonably good-quality audio file only requires delivery at 128 kbps. Figure 2.09 shows a simple schematic diagram of the components involved in the streaming.

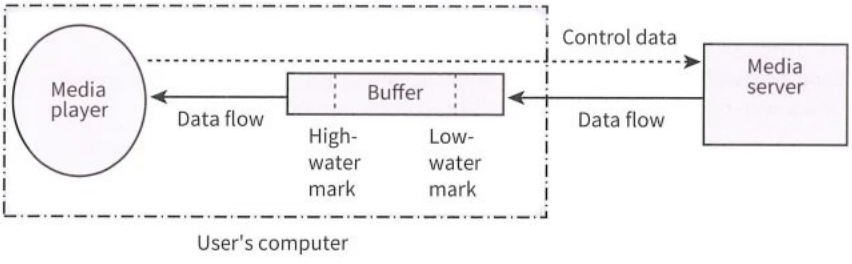


Figure 2.09 Schematic diagram of bit streaming

KEY TERMS

On-demand: when the bit stream content is transmitted at a time chosen by the user

Real-time: when the bit stream content is transmitted as it is produced

Bit rate: the number of bits transmitted per second

The buffer must deliver the data to the user, at the correct bit rate for the media being used. Data which is sent into the buffer should be sent at a higher rate to allow for unexpected delays. The media player continuously monitors how full the buffer is and controls the bit rate in relation to the defined high- and low-water marks. It is essential to have a buffer size that is sufficiently large for it never to get filled.

The rate of transmission to the buffer is limited by the bandwidth of the network connection. For a connection via a PSTN, a broadband link is essential. For good-quality movie presentation the broadband requirement is about 2.5 Mbps. Because this will not be available for all users it is often the practice that an individual video is made available at different levels of compression. The most highly compressed version will be the poorest quality but the bit rate may be sufficiently low for a reasonable presentation with a relatively low bandwidth Internet connection.

TASK 2.02

Consider a bit-streaming scenario for a video where the following values apply:

- the buffer size is 1 MiB
- the low-water mark is set at 100 KiB
- the high-water mark is set at 900 KiB
- the incoming data rate is 1 Mbps
- the video display rate is 300 Kbps.

Assume that the video is playing and that the buffer content has dropped to the low-water mark. The media player sets the controls for data input to begin again.

Calculate the amount of data that will be input to the buffer in two seconds and the amount of data that will be removed from the buffer in the same time period.

Repeat the calculation for 4, 6, 8, 10 and 12 seconds.

From this data, estimate when the buffer will have filled up to the high-water mark.

Assuming that the incoming transmission is halted at this time, calculate how long it will be before the buffer content has again fallen to the low-water mark level.

2.08 IP addressing

The Internet requires technical protocols to function. A protocol suite called TCP/IP is used as a standard (see Chapter 17). One aspect of this is IP addressing, which is used to define from where and to where data is being transmitted.

IPv4 addressing

Currently the Internet uses Internet Protocol version 4 (IPv4) addressing. IPv4 was devised in the late 1970s, before the invention of the PC and the mobile phone. IPv4 provides for a large but limited number of addresses for devices, which is no longer enough to cover all the devices expected to use the Internet in future.

The IPv4 addressing scheme is based on 32 bits (four bytes) being used to define an **IPv4 address**. It is worth putting this into context. The 32 bits allow 2^{32} different addresses. For big numbers like this it is worth remembering that 2^{10} is approximately 1000 in denary so the



KEY TERM

IPv4 address: a 32-bit long, hierarchical address of a device on the Internet